Course Name: Geographic Information System

Week 02 - Assignment 02 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

In a real-world model, the attributes are linked to _____.

a) Map symbols
b) Data quality
c) Database
d) Machine code

Answer: (c)

Detailed Solution:

In a real-world GIS model, attributes are linked to a database because databases store descriptive information (such as name, population, or land use) associated with spatial features shown on maps.

QUESTION 2

Ranking of an attribute is given by _____ scale.

a) Nominal
b) Ratio
c) Ordinal
d) Interval

Answer: (c)

Detailed Solution:

The ordinal scale is used for ranking attributes because it arranges data in a meaningful order or hierarchy, such as low, medium, and high, without specifying exact differences between them.

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QUESTION 3

An electrical power cable can be represented appropriately using _____ graphical feature.

a) Polygons
b) Lines
c) Points
d) All of the above

Answer: (b)

Detailed Solution:

An electrical power cable is best represented using a line feature because it has length and connectivity but negligible width when mapped in GIS.

QUESTION 4

Terrain of a place is categorised as _____ phenomena.

a) Discrete
b) Static
c) Fuzzy
d) Continuous

Answer: (d)

Detailed Solution:

Terrain is considered a continuous phenomenon because elevation and landforms change gradually over space rather than existing as separate isolated features.

QUESTION 5

Which of the following activity in a GIS routine is tedious and expensive?

a) Data processing
b) Data collection
c) Data analysis
d) Data manipulation

Answer: (b)

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Detailed Solution:

Data collection is the most tedious and expensive activity in GIS because gathering accurate spatial and attribute data from surveys, satellites, GPS, or fieldwork requires significant time, effort, and resources.

QUESTION 6

While dealing with a raster data, one of the key considerations is_____.

a) Resolution
b) Time
c) Attribute
d) Expenditure

Answer: (a)

Detailed Solution:

In raster data, resolution is a key consideration because it determines the size of each pixel and the level of detail represented in the dataset. Higher resolution provides more detail but requires larger storage and processing capacity.

QUESTION 7

Every sample has equal probability to be chosen in _____ sampling.

a) Stratified
b) Systematic
c) Random
d) Symmetric

Answer: (c)

Detailed Solution:

In random sampling, every sample has an equal probability of being selected. This reduces bias and ensures fair representation of the population.

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QUESTION 8

Digitizing is a method adopted to generate _____ data.

a) Primary
b) Vector
c) Raster
d) All of the above

Answer: (b)

Detailed Solution:

Digitizing is mainly used to generate vector data by converting maps, drawings, or images into digital points, lines, and polygons within a GIS environment.

QUESTION 9

Migration and housing data are examples of _____ data.

a) Attribute
b) Framework
c) Topographic
d) Socio-economic

Answer: (d)

Detailed Solution:

Migration and housing information are examples of socio-economic data because they describe social and economic characteristics of a population.

QUESTION 10

Dangles are _____ errors that occur during vector data input.

a) Systematic
b) Random
c) Digitizing
d) Scanning

Answer: (c)

Detailed Solution:

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Dangles are digitizing errors that occur during vector data input when lines fail to connect properly, such as overshoots or undershoots in network mapping.

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